ASSESSMENT CO1-AT1 • OOAD (CSA1111)

Object-Oriented Analysis & Design: Concept Maps

Comprehensive solutions covering OOP pillars, lifecycle workflows, object coupling, SOLID architecture, UML modeling, and pipeline troubleshooting.

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QUESTION 1 • CONCEPT MAP RECONSTRUCTION

Core Object-Oriented Pillars

Class → Object → **Encapsulation** → Inheritance → **Abstraction** → Polymorphism

Encapsulation

Bundles data and methods into isolated units.

Outcome: Guarantees data security and integrity.

Abstraction

Hides unnecessary details and exposes essential interfaces.

Outcome: Enforces simplicity and modular maintenance.

QUESTION 2 • LIFECYCLE EXECUTION

Software Development Life Cycle

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
Requirements	**Design**	**Coding**	**Testing**	**Deploy**	**Maintain**
Stakeholder needs	System models	Implementation	Verification	Release build	Evolution

Structured Progression

Transforms abstract user needs into validated production code.

Early Error Detection

Preempts expensive downstream failures during design reviews.

Architectural Modularity

Isolates concerns for decoupled deployment and scaling.

QUESTION 3 • DOMAIN MODELING

Object Relationships & Coupling

LEVEL 1 • LOOSE

Association

Basic operational peer link between independent objects.

Independent lifecycles.

LEVEL 2 • MEDIUM

Aggregation

Weak "Has-A" relationship where parts survive the parent.

Non-exclusive ownership.

LEVEL 3 • STRICT

Composition

Strong ownership where parts perish with the container.

Coupled lifecycle binding.

QUESTION 4 • ARCHITECTURAL RIGOR

Essential Design Principles

1. Modularity

Partitions monoliths into cohesive subsystems.

2. Single Responsibility (SRP)

One class has one reason to change.

3. Open / Closed Principle (OCP)

Open for extension, closed for modification.

SYSTEM IMPACT

Software Resilience

- Minimizes regression defects during upgrades.
- Eliminates structural coupling between layers.
- Accelerates team velocity and parallel testing.

QUESTION 5 • STANDARDIZED NOTATIONS

UML Diagrams & Perspectives

Use Case

User interaction and scope boundaries.

Class Diagram

Static structures and object associations.

Sequence

Chronological message interchange flow.

Activity

Operational decision workflows.

QUESTION 6 • FAILURE ANALYSIS

Implementation Defect Propagation

Requirement ✓ → Design ✓ → Coding ✗ (Failure) → Testing ✗ → Deployment ✗

Cascade Mechanism

Coding defects corrupt build integrity.

Testing halts and deployment aborts.

Remediation Plan

- Patch source logic at coding layer.
- Re-execute test suites prior to deployment.

ACADEMIC SUBMISSION COMPLETE

OOAD (CSA1111) Solutions

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